omeprazole (Rx, OTC)

Brand and Other Names: Prilosec, Prilosec OTC, OmepraCare, OmepraCare DR
Classes: Proton Pump Inhibitors

Dosing & Uses

ADULT

Dosage Forms & Strengths

packet

- 2.5mg
- 10mg

tablet, delayed release

- 20mg (Prilosec OTC, OmepraCare)

capsule, delayed release

- 10mg (generic)
- 20mg (OmepraCare DR, generic)
- 40mg (generic)

oral disintegrating tablets

- 20mg

Duodenal Ulcer

20 mg PO qDay for 4-8 weeks

Safety and efficacy of omeprazole for maintenance treatment past 1 year not established

Helicobacter Pylori Infection

Various regimens exist of PPIs combined with antibiotics, an example is listed below

20 mg PO q12hr for 10 days, WITH

Amoxicillin 1000 mg PO q12hr, AND

Clarithromycin 500 mg PO q12hr for 10-14 days

Dosing considerations

- This regimen is available as a prepackaged 10-day supply of omeprazole, amoxicillin, and clarithromycin from Dava Pharms Inc, for eradication of H pylori

Gastric Ulcer

40 mg PO qDay for 4-8 weeks

GERD

20 mg PO qDay for 4 weeks

Erosive Esophagitis

20 mg PO qDay for 4-8 weeks

Maintenance: 20 mg PO qDay for up to 1 year

Hypersecretory Conditions (eg, Zollinger-Ellison Syndrome)

60 mg PO qDay (initial) up to 360 mg/day divided q8hr PO

If dose >80 mg, divide it

Dosage Modifications

Hepatic impairment: Not studied; expert analysis recommends a reduction in dose, especially for maintenance of healing of erosive esophagitis

Renal impairment: Dose adjustments not necessary

PEDIATRIC

Dosage Forms & Strengths

packet

- 2.5mg
- 10mg

tablet, delayed release

- 20mg (Prilosec OTC, OmepraCare)

capsule, delayed release

- 10mg (generic)
- 20mg (OmepraCare DR, generic)
- 40mg (generic)

oral disintegrating tablets

- 20mg

GERD

Indicated for treatment of GERD

<1 year: Safety and efficacy not established

5-10 kg: 5 mg PO qDay

10-20 kg: 10 mg PO qDay

>20 kg: 20 mg PO qDay

Erosive Esophagitis

Indicated for treatment and to maintain healing of erosive esophagitis caused by acid-mediated GERD

Treatment

- <1 month: Safety and efficacy not established
- Aged 1 month to <1 year
  - 3 to <5 kg: 2.5 mg qDay
  - 5 to <10 kg: 5 mg qDay
  - ≥10 kg: 10 mg qDay
  - May treat for up to 6 weeks
- Aged 1-16 years
  - 5 to <10 kg: 5 mg PO qDay
  - 10 to <20 kg: 10 mg PO qDay
  - ≥20 kg: 20 mg PO qDay

- May treat for 4-8 weeks

Maintenance of healing

- <1 year: Safety and efficacy not established
- ≥1 year: Controlled trials for maintenance do not extend beyond 12 months

Neonates (Off-label)

Refractory duodenal ulcer or reflux esophagitis: 0.5-1.5 mg/kg PO qDay for up to 8 weeks

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Interactions

Interaction Checker

Enter a drug name to check for any interactions. Type a drug, OTC, or herbal and omeprazole
No Interactions Found
Interactions Found

Contraindicated

Serious

Significant - Monitor Closely

Minor

All Interactions Sort By: Severity

Contraindicated (2)

- nelfinavir
  omeprazole decreases levels of nelfinavir by unspecified interaction mechanism. Contraindicated. Coadministration may lead to loss of nelfinavir virologic response and development of resistance; mechanism may be CYP2C19 inhibition of nelfinavir conversion to active M8 metabolite, and also PPIs decreasing gastric pH resulting in decreased nelfinavir absorption.

- rilpivirine
  omeprazole decreases levels of rilpivirine by increasing gastric pH. Applies only to oral form of both agents. Contraindicated. Concurrent use may cause treatment failure and/or the development of rilpivirine or NNRTI resistance owing to decreased levels.

Serious (41)

- abrocitinib

omeprazole will increase the level or effect of abrocitinib by decreasing metabolism. Avoid or Use Alternate Drug. Abrocitinib is a CYP2C9 and CYP2C19 substrate. Drugs that are moderate-to-strong inhibitors of both CYP2C9 and CYP2C19 increase systemic exposure of abrocitinib and its active metabolites, which may increase adverse effects.

- acalabrutinib
  omeprazole decreases levels of acalabrutinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Acalabrutinib solubility decreases with increasing gastric pH. Due to the long-lasting effect of PPIs, separation of doses may not eliminate the interaction.

- apalutamide
  apalutamide will decrease the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Coadministration of apalutamide, a strong CYP2C19 inducer, with drugs that are CYP2C19 substrates can result in lower exposure to these medications. Avoid or substitute another drug for these medications when possible. Evaluate for loss of therapeutic effect if medication must be coadministered.

  apalutamide will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Coadministration of apalutamide, a strong CYP3A4 inducer, with drugs that are CYP3A4 substrates can result in lower exposure to these medications. Avoid or substitute another drug for these medications when possible. Evaluate for loss of therapeutic effect if medication must be coadministered. Adjust dose according to prescribing information if needed.

- atazanavir
  omeprazole will decrease the level or effect of atazanavir by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Atazanavir solubility decreases as pH increases. Substantially reduced plasma concentrations of atazanavir are expected if PPIs are coadministered. PPI dose should not exceed a dose comparable to omeprazole 20 mg and must be taken ~12 h before atazanavir/ritonavir in treatment naive-patients. PPIs are not recommended in treatment-experienced taking atazanavir.

- carbamazepine
  carbamazepine will decrease the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug.

- ceritinib
  omeprazole decreases effects of ceritinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- cilostazol
  omeprazole will increase the level or effect of cilostazol by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug.

- clopidogrel
  omeprazole decreases effects of clopidogrel by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Clopidogrel efficacy may be reduced by drugs that inhibit CYP2C19. Inhibition of platelet aggregation by clopidogrel is

entirely due to an active metabolite. Clopidogrel is metabolized to this active metabolite in part by CYP2C19. .

- dacomitinib
  omeprazole will increase the level or effect of dacomitinib by unspecified interaction mechanism. Avoid or Use Alternate Drug. Concomitant use with a PPI decreases dacomitinib concentrations, which may reduce dacomitinib efficacy. Avoid use of PPIs with dacomitinib. As an alternative to PPIs, use locally-acting antacids or an H2-receptor antagonist. Administer at least 6 hours before or 10 hours after taking an H2-receptor antagonist.

- dasatinib
  omeprazole will decrease the level or effect of dasatinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- defactinib
  omeprazole decreases levels of defactinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Defactinib exposure decreases by 79% following concomitant use of multiple doses of omeprazole 40 mg PO daily. If concomitant use with PPIs cannot be avoided, separate administration by at least 2 hr.

- digoxin
  omeprazole will increase the level or effect of digoxin by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- enzalutamide
  enzalutamide will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug.

- erlotinib
  omeprazole decreases levels of erlotinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration of proton pump inhibitors and erlotinib may decrease erlotinib exposure and reduce its efficacy. Consider alternate acid reducing agents.

- fedratinib
  omeprazole will increase the level or effect of fedratinib by Other (see comment). Avoid or Use Alternate Drug. Avoid coadministration of fedratinib (a CYP3A4 and CYP2C19 substrate) with dual CYP3A4 and CYP2C19 inhibitor. Effect of coadministration of a dual CYP3A4 and CYP2C19 inhibitor with fedratinib has not been studied.

- fluvoxamine
  fluvoxamine will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug.

- idelalisib
  idelalisib will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Idelalisib is a strong CYP3A inhibitor; avoid coadministration with sensitive CYP3A substrates

- infigratinib (DSC)
  omeprazole will decrease the level or effect of infigratinib (DSC) by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- isoniazid
  isoniazid will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug.

- itraconazole
  omeprazole will decrease the level or effect of itraconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- ivosidenib
  ivosidenib will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Avoid coadministration of sensitive CYP3A4 substrates with ivosidenib or replace with alternative therapies. If coadministration is unavoidable, monitor patients for loss of therapeutic effect of these drugs.

- ketoconazole
  omeprazole will decrease the level or effect of ketoconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- levoketoconazole
  omeprazole will decrease the level or effect of levoketoconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- lonafarnib
  omeprazole will increase the level or effect of lonafarnib by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. If coadministration of lonafarnib (a sensitive CYP3A substrate) with weak CYP3A inhibitors is unavoidable, reduce to, or continue lonafarnib at starting dose. Closely monitor for arrhythmias and events (eg, syncope, heart palpitations) since lonafarnib effect on QT interval is unknown.

  lonafarnib will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Avoid or Use Alternate Drug. Lonafarnib may increase the AUC and peak concentration of CYP2C19 substrates. If coadministration unavoidable, monitor for adverse reactions and reduce the CYP2C19 substrate dose in accordance with its approved product labeling.

- mesalamine
  omeprazole decreases effects of mesalamine by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Applies only to sustained release dosage form.

- neratinib
  omeprazole will decrease the level or effect of neratinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- nilotinib
  omeprazole decreases levels of nilotinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. As an alternative to PPIs,

use H2-blockers ~10 hr before or ~2 hr after nilotinib, or use antacids ~2 hr before or ~2 hr after nilotinib (does not apply if using ODT nilotinib).

- nisoldipine
  omeprazole will increase the level or effect of nisoldipine by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- pazopanib
  omeprazole will decrease the level or effect of pazopanib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Avoid coadministration of pazopanib with drugs that raise gastric pH; consider short-acting antacids in place of PPIs and H2 antagonists; separate antacid and pazopanib dosing by several hours

- pexidartinib
  omeprazole will decrease the level or effect of pexidartinib by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Avoid coadministration of proton pump inhibitors (PPIs) with pexidartinib. Use H2-receptor antagonists or antacids if needed. When using alternatives to PPIs, administer pexidartinib 2 hr before or after taking locally-acting antacids OR administer pexidartinib at least 2 hr before or 10 hr after taking an H2-receptor antagonist.

- phenobarbital
  phenobarbital will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug.

- rifampin
  rifampin will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug.

- rifapentine
  rifapentine will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug.

- rilzabrutinib
  omeprazole decreases levels of rilzabrutinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration may decrease rilzabrutinib exposure, resulting in reduced efficacy.

- secretin
  omeprazole, secretin. Other (see comment). Avoid or Use Alternate Drug. Comment: Concomitant use of PPIs may cause a hyperresponse in gastrin secretion in response to stimulation testing with secretin, falsely suggesting gastrinoma. The time it takes for serum gastrin concentrations to return to baseline following discontinuation of PPIs is specific to the individual PPI. Temporarily stop omeprazole treatment at least 14 days before assessing to allow gastrin levels to return to baseline.

- sofosbuvir/velpatasvir
  omeprazole will decrease the level or effect of sofosbuvir/velpatasvir by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Velpatasvir solubility decreases as gastric pH increases (practically insoluble at pH

>5). Coadministration of sofosbuvir/velpatasvir with omeprazole or other PPIs is not recommended. If considered medically necessary, give sofosbuvir/velpatasvir with food 4 hr before omeprazole 20 mg. Use with other PPIs has not been studied.

- sotorasib
  omeprazole will decrease the level or effect of sotorasib by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug. If use with an acid-reducing agent cannot be avoided, administer sotorasib 4 hr before or 10 hr after administration of a locally-acting antacid.

- sparsentan
  omeprazole decreases effects of sparsentan by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Proton pump inhibitors may decrease sparsentan exposure which may reduce efficacy of sparsentan.

- taletrectinib
  omeprazole decreases levels of taletrectinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Coadministration may decrease taletrectinib exposure and reduce its efficacy.

- tucatinib
  tucatinib will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. Avoid concomitant use of tucatinib with CYP3A substrates, where minimal concentration changes may lead to serious or life-threatening toxicities. If unavoidable, reduce CYP3A substrate dose according to product labeling.

- ziftomenib
  omeprazole decreases levels of ziftomenib by unspecified interaction mechanism. Avoid or Use Alternate Drug. Proton pump inhibitors decrease zifomenib peak plasma concentations by 70%.

Monitor Closely (94)

- abrocitinib
  omeprazole will increase the level or effect of abrocitinib by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Start abrocitinib 50 mg qDay when coadministered with CYP2C19 inhibitors. If adequate response not achieved after 12 weeks, may increase to 100 mg qDay. Discontinue if inadequate response after dosage increase.

- amphetamine
  omeprazole, amphetamine. Other (see comment). Use Caution/Monitor. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .

- ampicillin
  omeprazole will decrease the level or effect of ampicillin by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- armodafinil

armodafinil will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- avapritinib
  omeprazole will increase the level or effect of avapritinib by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- axitinib
  omeprazole increases levels of axitinib by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- belumosudil
  omeprazole will decrease the level or effect of belumosudil by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Increase belumosudil dosage to 200 mg PO BID when coadministered with proton pump inhibitors.

- belzutifan
  omeprazole will increase the level or effect of belzutifan by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Belzutifan is a CYP2C19 substrate. Coadministration with CYP2C19 inhibitors may increase incidence or severity of adverse effects. Monitor for anemia and hypoxia and reduce belzutifan dose as recommended.

- bortezomib
  bortezomib will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- bosutinib
  omeprazole decreases levels of bosutinib by Other (see comment). Use Caution/Monitor. Comment: PPIs may decrease bosutinib concentration by ~45%; bosutinib displays pH-dependent solubility.

- budesonide
  omeprazole decreases effects of budesonide by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Enteric-coated budesonide dissolves at pH >5.5. Also, dissolution of extended-release budesonide tablets is pH dependent. Coadministration with drugs that increase gastric pH may cause these budesonide products to prematurely dissolve, and possibly affect release properties and absorption of the drug in the duodenum.

- butalbital
  butalbital will decrease the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- cannabidiol
  omeprazole will increase the level or effect of cannabidiol by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the cannabidiol dose when coadministered with a strong CYP2C19 inhibitor.

  cannabidiol will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the dose of sensitive CYP2C19 substrates, as clinically appropriate, when coadministered with cannabidiol.

- capecitabine
  omeprazole decreases effects of capecitabine by unknown mechanism. Use Caution/Monitor. Retrospective data suggest elevated gastric pH caused by PPI use may impair capecitabine tablet dissolution and/or reduce absorption.

- carbamazepine
  carbamazepine will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- carbonyl iron
  omeprazole will decrease the level or effect of carbonyl iron by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- cefpodoxime
  omeprazole decreases effects of cefpodoxime by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- cefuroxime
  omeprazole decreases effects of cefuroxime by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- cenobamate
  cenobamate will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Increase dose of CYP3A4 substrate, as needed, when coadministered with cenobamate.

  cenobamate will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider a dose reduction of CYP2C19 substrates, as clinically appropriate, when used concomitantly with cenobamate.

- ciprofloxacin
  omeprazole will decrease the level or effect of ciprofloxacin by unknown mechanism. Use Caution/Monitor. Absorption of the ciprofloxacin ER tablet was slightly diminished (20%) when coadministered with omeprazole.

- citalopram
  omeprazole will increase the level or effect of citalopram by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Citalopram 20 mg/day is the maximum recommended dose for patients taking CYP2C19 inhibitors because of the risk of QT prolongation.

- clobazam
  omeprazole will increase the level or effect of clobazam by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Dosage adjustment may be required; CYP2C19 inhibitors may result in increased exposure to N-desmethylclobazam (active metabolite).

- cobicistat

cobicistat will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- conivaptan
  conivaptan will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- crizotinib
  omeprazole will decrease the level or effect of crizotinib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Drugs that elevate the gastric pH may decrease the solubility of crizotinib and subsequently reduce its bioavailability. However, no formal studies have been conducted.

- cyclosporine
  omeprazole, cyclosporine. Either increases toxicity of the other by Other (see comment). Use Caution/Monitor. Comment: Long-term use of PPIs may cause hypomagnesemia and increase this risk when coadministered with drugs that may also decrease magnesium levels.

- dabrafenib
  omeprazole will decrease the level or effect of dabrafenib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Drugs that alter upper GI tract pH (eg, PPIs, H2-blockers, antacids) may decrease dabrafenib solubility and reduce its bioavailability

  dabrafenib will decrease the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Use alternative if available

- dextroamphetamine
  omeprazole, dextroamphetamine. Other (see comment). Use Caution/Monitor. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .

- diazepam buccal
  omeprazole will increase the level or effect of diazepam buccal by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Strong or moderate CYP2C19 inhibitors may decrease rate of diazepam elimination, thereby increasing adverse reactions to diazepam.

- diazepam intranasal
  omeprazole will increase the level or effect of diazepam intranasal by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Strong or moderate CYP2C19 inhibitors may decrease rate of diazepam elimination, thereby increasing adverse reactions to diazepam.

- dichlorphenamide
  dichlorphenamide and omeprazole both decrease serum potassium. Use Caution/Monitor.

- digoxin
  omeprazole increases toxicity of digoxin by Other (see comment). Use Caution/Monitor. Comment: Prolonged use of PPIs may cause hypomagnesemia and

increase risk for digoxin toxicity.

- dordaviprone
  dordaviprone and omeprazole both increase QTc interval. Use Caution/Monitor. Increase the frequency of ECG and electrolyte monitoring.

- efavirenz
  efavirenz will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

  efavirenz will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- elagolix
  elagolix will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Elagolix is a weak inhibitor of CYP2C19. No dose adjustments are needed for omeprazole with 40 mg/day or less. Consider omeprazole dose reduction with higher omeprazole doses (eg, dose for Zollinger-Ellison) if coadministered with elagolix.

  elagolix decreases levels of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Elagolix is a weak-to-moderate CYP3A4 inducer. Monitor CYP3A substrates if coadministered. Consider increasing CYP3A substrate dose if needed.

- eluxadoline
  omeprazole increases levels of eluxadoline by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. As a precautionary measure due to incomplete information on the metabolism of eluxadoline, use caution when coadministered with strong CYP2C19 inhibitors.

- encorafenib
  encorafenib, omeprazole. affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Encorafenib both inhibits and induces CYP3A4 at clinically relevant plasma concentrations. Coadministration of encorafenib with sensitive CYP3A4 substrates may result in increased toxicity or decreased efficacy of these agents.

- escitalopram
  omeprazole will increase the level or effect of escitalopram by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- eslicarbazepine acetate
  eslicarbazepine acetate will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- etravirine
  etravirine will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- fedratinib
  fedratinib will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Adjust dose

of drugs that are CYP3A4 substrates as necessary.

- ferric gluconate
  omeprazole will decrease the level or effect of ferric gluconate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ferric maltol
  omeprazole will decrease the level or effect of ferric maltol by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ferrous fumarate
  omeprazole will decrease the level or effect of ferrous fumarate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ferrous gluconate
  omeprazole will decrease the level or effect of ferrous gluconate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ferrous sulfate
  omeprazole will decrease the level or effect of ferrous sulfate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- fexinidazole
  fexinidazole will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- finerenone
  omeprazole will increase the level or effect of finerenone by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Monitor serum potassium during initiation and dosage adjustment of either finererone or weak CYP3A4 inhibitors. Adjust finererone dosage as needed.

- flibanserin
  omeprazole will increase the level or effect of flibanserin by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Coadministration of flibanserin with strong CYP2C19 inhibitors may increase flibanserin exposure and increase the risk of hypotension, syncope, and CNS depression.

  omeprazole will increase the level or effect of flibanserin by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Increased flibanserin adverse effects may occur if coadministered with multiple weak CYP3A4 inhibitors.

- fluconazole
  fluconazole will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- fosamprenavir
  omeprazole will decrease the level or effect of fosamprenavir by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- fosphenytoin

fosphenytoin will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- gefitinib
  omeprazole decreases levels of gefitinib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Avoid coadministration of gefitinib with PPIs if possible. If treatment with a PPI is required, separate gefitinib and PPI doses by 12 hr.

- imipramine
  omeprazole will increase the level or effect of imipramine by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- iron dextran complex
  omeprazole will decrease the level or effect of iron dextran complex by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- iron sucrose
  omeprazole will decrease the level or effect of iron sucrose by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- isavuconazonium sulfate
  omeprazole will increase the level or effect of isavuconazonium sulfate by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- istradefylline
  istradefylline will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Istradefylline 40 mg/day increased peak levels and AUC of CYP3A4 substrates in clinical trials. This effect was not observed with istradefylline 20 mg/day. Consider dose reduction of sensitive CYP3A4 substrates.

- ledipasvir/sofosbuvir
  omeprazole decreases levels of ledipasvir/sofosbuvir by Other (see comment). Use Caution/Monitor. Comment: Ledipasvir solubility decreases as pH increases; drugs that increase gastric pH are expected to decrease levels of ledipasvir; proton-pump inhibitor doses comparable to omeprazole <20 mg can be administered simultaneously with ledipasvir/sofosbuvir under fasted conditions.

- lemborexant
  omeprazole will increase the level or effect of lemborexant by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Lower nightly dose of lemborexant recommended if coadministered with weak CYP3A4 inhibitors. See drug monograph for specific dosage modification.

- letermovir
  letermovir increases levels of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Monitor and dose adjustment may be necessary.

- lumacaftor/ivacaftor
  lumacaftor/ivacaftor, omeprazole. affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. In vitro studies suggest that lumacaftor may induce and

ivacaftor may inhibit CYP2C19 substrates. .

lumacaftor/ivacaftor will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Lumacaftor/ivacaftor is a strong CYP3A4 inhibitor and also has the potential to induce CYP2C19 and both induce and inhibitor P-gp.

- mavacamten
  omeprazole increases levels of mavacamten by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Modify mavacamten dose if coadministered with weak or moderate CYP2C19 inhibitors.

- methotrexate
  omeprazole increases levels of methotrexate by decreasing renal clearance. Use Caution/Monitor. Temporary withdrawal of PPI may be considered in some patients.

- methylphenidate
  omeprazole decreases effects of methylphenidate by enhancing GI absorption. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Since the characteristics of methylphenidate extended release capsules (Ritalin LA) are pH dependent, coadministration of antacids or acid suppressants could alter the release of methylphenidate. Consider separating the administration of the antacid and the methylphenidate extended-release capsules may be avoided.

- midazolam intranasal
  omeprazole will increase the level or effect of midazolam intranasal by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor. Coadministration of mild CYP3A4 inhibitors with midazolam intranasal may cause higher midazolam systemic exposure, which may prolong sedation.

- mitapivat
  mitapivat decreases levels of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Monitor patients for loss of therapeutic effect if taking sensitive CYP2C19 substrates. .

- modafinil
  modafinil will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- mycophenolate
  omeprazole will decrease the level or effect of mycophenolate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Potential interaction applies to mycophenolate mofetil. Enteric coated mycophenolate sodium formulation is less sensitive to this interaction. Clinical significance unclear.

- ombitasvir/paritaprevir/ritonavir & dasabuvir (DSC)
  ombitasvir/paritaprevir/ritonavir & dasabuvir (DSC) will decrease the level or effect of omeprazole by unspecified interaction mechanism. Use Caution/Monitor. Monitor for decreased omeprazole efficacy; consider increasing omeprazole dose in patients whose symptoms are not well controlled (not to exceed 40 mg/day)

- oxcarbazepine

oxcarbazepine will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- paltusotine
  omeprazole decreases levels of paltusotine by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Coadministration demonstrated a dose-dependent decrease in paltusotine exposure owing to decreased paltusotine solubility at elevated gastric pH. May require increased paltusotine dose. Avoid coadministration if patient is already taking paltusotine 60 mg.

- pentobarbital
  pentobarbital will decrease the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- phenobarbital
  phenobarbital will decrease the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- phenytoin
  phenytoin will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- polysaccharide iron
  omeprazole will decrease the level or effect of polysaccharide iron by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- posaconazole
  omeprazole will decrease the level or effect of posaconazole by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

  posaconazole will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- primidone
  primidone will decrease the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

  primidone will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- revumenib
  revumenib and omeprazole both increase QTc interval. Use Caution/Monitor. ECG monitoring recommended if concomitant use can?t be avoided and withhold therapy if QTc interval is >480 msec; restart therapy after QTc interval returns to <480 msec

- rifampin
  rifampin will decrease the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- riociguat

omeprazole will decrease the level or effect of riociguat by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ritonavir
  ritonavir will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- rose hips
  omeprazole will decrease the level or effect of rose hips by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- sparsentan
  sparsentan will decrease the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Sparsentan (a CYP2C19 inducer) decreases exposure of CYP2C19 substrates and reduces efficacy related to these substrates.

- St John's Wort
  St John's Wort will decrease the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- stiripentol
  stiripentol, omeprazole. affecting hepatic/intestinal enzyme CYP3A4 metabolism. Modify Therapy/Monitor Closely. Stiripentol is a CYP3A4 inhibitor and inducer. Monitor CYP3A4 substrates coadministered with stiripentol for increased or decreased effects. CYP3A4 substrates may require dosage adjustment.

  stiripentol will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Modify Therapy/Monitor Closely. Consider reducing the dose of CYP2C19 substrates, if adverse reactions are experienced when administered concomitantly with stiripentol.

- tacrolimus
  omeprazole will increase the level or effect of tacrolimus by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. Concomitant administration may increase tacrolimus whole blood concentrations, particularly in intermediate or poor metabolizers of CYP2C19

- taletrectinib
  taletrectinib and omeprazole both increase QTc interval. Use Caution/Monitor. Concomitant use may increase the risk of QTc interval prolongation. If concomitant use is unavoidable, adjust the frequency of ECG and electrolyte monitoring.

- tinidazole
  omeprazole will increase the level or effect of tinidazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- tofacitinib
  omeprazole increases levels of tofacitinib by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor. No specific dose adjustment recommended when tofacitinib coadministered with potent CYP2C19 inhibitors; decrease tofacitinib

dose if coadministered with both moderate CYP3A4 and potent CYP2C19 inhibitors
.

- triclabendazole
  triclabendazole will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

- voriconazole
  voriconazole will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C19 metabolism. Use Caution/Monitor.

  voriconazole will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Use Caution/Monitor.

- warfarin
  omeprazole will increase the level or effect of warfarin by Other (see comment). Use Caution/Monitor. Warfarin's less potent R-enantiomer is metabolized in part by CYP3A4 (and also CYP1A2 and CYP2C19). Monitor INR more frequently if coadministered with inhibitors of these isoenzymes and adjust warfarin dose if needed.

- ziftomenib
  omeprazole, ziftomenib. QTc interval. Use Caution/Monitor. Caution if ziftomenib is coadministered with drugs that exacerbated risk of QT prolongation (eg, hypokalemia, orthostatic hypotension, inhibits metabolism of ziftomenib).

Minor (50)

- acetazolamide
  acetazolamide will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- alprazolam
  omeprazole increases levels of alprazolam by decreasing metabolism. Minor/Significance Unknown.

- ambrisentan
  omeprazole will increase the level or effect of ambrisentan by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- anastrozole
  anastrozole will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- bismuth subsalicylate
  omeprazole increases levels of bismuth subsalicylate by enhancing GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.

- blessed thistle
  blessed thistle decreases effects of omeprazole by pharmacodynamic antagonism. Minor/Significance Unknown. Theoretical interaction.

- bosentan
  bosentan will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- carbamazepine
  omeprazole increases levels of carbamazepine by decreasing metabolism. Minor/Significance Unknown. Monitor plasma levels when used concomitantly.

- chlordiazepoxide
  omeprazole increases levels of chlordiazepoxide by decreasing metabolism. Minor/Significance Unknown.

- clomipramine
  omeprazole will increase the level or effect of clomipramine by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- clonazepam
  omeprazole increases levels of clonazepam by decreasing metabolism. Minor/Significance Unknown.

- clorazepate
  omeprazole increases levels of clorazepate by decreasing metabolism. Minor/Significance Unknown.

- cyanocobalamin
  omeprazole decreases levels of cyanocobalamin by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.

- cyclophosphamide
  cyclophosphamide will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- danazol
  danazol will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- darifenacin
  darifenacin will decrease the level or effect of omeprazole by Other (see comment). Minor/Significance Unknown. Effectiveness of proton pump inhibitors may be decreased theoretically if administered with other antisecretory agents

- devil's claw
  devil's claw decreases effects of omeprazole by pharmacodynamic antagonism. Minor/Significance Unknown.

- dexamethasone
  dexamethasone will decrease the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- diazepam
  omeprazole will increase the level or effect of diazepam by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- dronedarone
  dronedarone will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- drospirenone
  drospirenone will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- elvitegravir/cobicistat/emtricitabine/tenofovir DF
  elvitegravir/cobicistat/emtricitabine/tenofovir DF, omeprazole. affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown. Based on drug interaction studies conducted with the components of Stribild, no clinically significant drug interactions have been either observed or are expected when coadministered with PPIs.

- eslicarbazepine acetate
  eslicarbazepine acetate decreases effects of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown. Monitor for GI symptoms; net increased or decreased effect on PPI action unclear due to opposing CYP450 actions.

- estazolam
  omeprazole increases levels of estazolam by decreasing metabolism. Minor/Significance Unknown.

- etravirine
  omeprazole will increase the level or effect of etravirine by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- ferric carboxymaltose
  omeprazole will decrease the level or effect of ferric carboxymaltose by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown.

- fluoxetine
  fluoxetine will increase the level or effect of omeprazole by affecting hepatic enzyme CYP2C9/10 metabolism. Minor/Significance Unknown.

- flurazepam
  omeprazole increases levels of flurazepam by decreasing metabolism. Minor/Significance Unknown.

- glipizide
  omeprazole will increase the level or effect of glipizide by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- larotrectinib
  larotrectinib will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- levoketoconazole
  levoketoconazole will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- levothyroxine
  omeprazole decreases levels of levothyroxine by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- liothyronine
  omeprazole decreases levels of liothyronine by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- liotrix
  omeprazole decreases levels of liotrix by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- lisdexamfetamine
  omeprazole, lisdexamfetamine. Other (see comment). Minor/Significance Unknown. Comment: Reduced gastric acidity caused by proton pump inhibitors decreases time to Tmax for amphetamine and dextroamphetamine. AUC was unaffected. .

- lorazepam
  omeprazole increases levels of lorazepam by decreasing metabolism. Minor/Significance Unknown.

- methamphetamine
  omeprazole decreases levels of methamphetamine by Other (see comment). Minor/Significance Unknown. Comment: Time to maximum concentration (Tmax) of amphetamine is decreased compared to when administered alone; monitor patients for changes in clinical effect and adjust therapy based on clinical response.

- midazolam
  omeprazole increases levels of midazolam by decreasing metabolism. Minor/Significance Unknown.

- mitotane
  mitotane decreases levels of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown. Mitotane is a strong inducer of cytochrome P-4503A4; monitor when coadministered with CYP3A4 substrates for possible dosage adjustments.

- ospemifene
  omeprazole increases levels of ospemifene by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

- phytoestrogens
  omeprazole decreases levels of phytoestrogens by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.

- ponatinib
  omeprazole decreases levels of ponatinib by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown.

- quazepam

omeprazole increases levels of quazepam by decreasing metabolism. Minor/Significance Unknown.

- ribociclib
  ribociclib will increase the level or effect of omeprazole by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- ruxolitinib
  omeprazole will increase the level or effect of ruxolitinib by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- ruxolitinib topical
  omeprazole will increase the level or effect of ruxolitinib topical by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Minor/Significance Unknown.

- theophylline
  omeprazole increases toxicity of theophylline by Other (see comment). Minor/Significance Unknown. Comment: Prolonged use of proton pump inhibitors can cause hypochlorhydria, which in turn causes peristalsis in small intestine to increase and peristalsis in the proximal colon to decrease; monitor for toxicity.

- thyroid desiccated
  omeprazole decreases levels of thyroid desiccated by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown. Conflicting evidence regarding this interaction exists.

- triazolam
  omeprazole increases levels of triazolam by decreasing metabolism. Minor/Significance Unknown.

- voriconazole
  omeprazole will increase the level or effect of voriconazole by affecting hepatic enzyme CYP2C19 metabolism. Minor/Significance Unknown.

Adverse Effects

1-10%

Headache (7%)

Abdominal pain (5%)

Diarrhea (4%)

Nausea (4%)

Vomiting (3%)

Flatulence (3%)

Dizziness (2%)

Upper respiratory infection (2%)

Acid regurgitation (2%)

Constipation (2%)

Rash (2%)

Cough (1%)

Frequency Not Defined

Fracture of bone, osteoporosis-related

Hepatotoxicity (rare)

Agranulocytosis

Anorexia

Gastric polyps

Hip fracture

Alopecia

Atrophic gastritis

Interstitial nephritis (rare)

Pancreatitis (rare)

Rhabdomyolysis

Taste perversion

Abnormal dreams

Toxic epidermal necrolysis (rare)

Postmarketing Reports

Bone fracture

Fundic gland polyps

Acute tubulointerstitial nephritis

Erectile dysfunction

Warnings

Contraindications

Hypersensitivity to omeprazole or other proton pump inhibitors (PPIs)

Cautions

PPIs are possibly associated with increased incidence of Clostridium difficile-associated diarrhea (CDAD); consider diagnosis of CDAD for patients taking PPIs who have diarrhea that does not improve

May require dosage reduction with liver disease

Cutaneous lupus erythematosus (CLE) and systemic lupus erythematosus (SLE) reported with PPIs; avoid using for longer than medically indicated; discontinue if signs or symptoms consistent with CLE or SLE are observed and refer patient to specialist

Shown to cause gastric carcinoid tumors in rats with increased doses, but risk in humans unconfirmed

Published observational studies suggest that PPI therapy may be associated with an increased risk for osteoporosis-related fractures of the hip, wrist, or spine, particularly with prolonged (>1 yr), high-dose therapy

Hypomagnesemia may occur with prolonged use (>1 year); adverse effects may result and include tetany, arrhythmias, and seizures; in 25% of cases reviewed, magnesium supplementation alone did not improve low serum magnesium levels and the PPI had to be discontinued

Decreased gastric acidity increases serum chromogranin A (CgA) levels and may cause false-positive diagnostic results for neuroendocrine tumors; temporarily discontinue PPIs before assessing CgA levels

Inhibits hepatic isoenzyme CYP2C19 and may alter metabolism of drugs that are CYP2C19 substrates

PPIs may decrease the efficacy of clopidogrel by reducing the formation of the active metabolite

Stop use and inform a healthcare professional if rash or joint pain develops

Daily long-term use (e.g., longer than 3 years) may lead to malabsorption or a deficiency of cyanocobalamin

Acute interstitial nephritis has been observed in patients taking PPIs

Relief of symptoms does not eliminate the possibility of a gastric malignancy

Therapy increases risk of Salmonella, Campylobacter, and other infections

May elevate and/or prolong serum concentrations of methotrexate and/or its metabolite when administered concomitantly with PPIs, possibly leading to toxicity; consider a temporary withdrawal of PPI therapy therapy with high dose methotrexate administration

PPI therapy is associated with increased risk of fundic gland polyp; risk increases with long-term use >1 year; patient may be asymptomatic; problem usually identified incidentally on endoscopy; use shortest duration of therapy appropriate to condition being treated

If patient taking a prescription drug, the patient should ask a doctor or a pharmacist whether acid reducers can be taken concomitantly with it

Therapy may cause severe skin reactions, including skin reddening, blisters, rash; if an allergic reaction occurs, stop use and seek medical help right away

Acute tubulointerstitial nephritis (TIN) reported in patients taking PPIs; may occur at any point during PPI therapy; patients may present with varying signs and symptoms from symptomatic hypersensitivity reactions to non-specific symptoms of decreased renal function (eg, malaise, nausea, anorexia); in reported case series, some patients were diagnosed on biopsy and in absence of extra-renal manifestations (eg, fever, rash or arthralgia); discontinue therapy and evaluate patients with suspected acute TIN

Pregnancy & Lactation

Pregnancy

There are no adequate and well-controlled studies in pregnant women; available epidemiologic data fail to demonstrate an increased risk of major congenital malformations or other adverse pregnancy outcomes with first trimester omeprazole use; reproduction studies in rats and rabbits resulted in dose-dependent embryo-lethality at omeprazole doses that were approximately 3.4 to 34 times an oral human dose of 40 mg (based on a body surface area for a 60 kg person)

Teratogenicity was not observed in animal reproduction studies with administration of oral esomeprazole (an enantiomer of omeprazole) magnesium in rats and rabbits during organogenesis with doses about 68 times and 42 times, respectively, an oral human dose of 40 mg esomeprazole or 40 mg omeprazole (based on body surface area for a 60 kg person); changes in bone morphology were observed in offspring of rats dosed through most of pregnancy and lactation at doses equal to or greater than approximately 34 times an oral human dose of 40 mg esomeprazole or 40 mg omeprazole; when maternal administration was confined to gestation only, there were no effects on bone physeal morphology in offspring at any age

Lactation

Limited data suggest omeprazole may be present in human milk; there are no clinical data on effects of omeprazole on breastfed infant or on milk production; developmental and health benefits of breastfeeding should be considered along with mother's clinical need for therapy and any potential adverse effects on breastfed infant from treatment or from underlying maternal condition

Pregnancy Categories

A: Generally acceptable. Controlled studies in pregnant women show no evidence of fetal risk.

B: May be acceptable. Either animal studies show no risk but human studies not available or animal studies showed minor risks and human studies done and showed no risk.

C: Use with caution if benefits outweigh risks. Animal studies show risk and human studies not available or neither animal nor human studies done.

D: Use in LIFE-THREATENING emergencies when no safer drug available. Positive evidence of human fetal risk.

X: Do not use in pregnancy. Risks involved outweigh potential benefits. Safer alternatives exist.

NA: Information not available.

Pharmacology

Mechanism of Action

PPI; binds to H+/K+-exchanging ATPase (proton pump) in gastric parietal cells, resulting in suppression of basal and stimulated acid secretion

Absorption

Bioavailability: 30-40%

Onset of action: 1 hr (antisecretory effect)

Duration: 73 hr

Peak plasma time: 0.5-3.5 hr

Peak response (PUD): 2 hr (initial); 5 days (peak)

Distribution

Protein bound: 95-96%

Vd: 0.39 L/kg

Metabolism

Metabolized extensively by hepatic CYP2C19; slow metabolizers are deficient in CYP2C19 enzyme system; plasma concentration can increase by 5-fold or higher in comparison with that found in persons with the enzyme

Metabolites: Hydroxyomeprazole, omeprazole sulfone, omeprazole sulfide (inactive)

Enzymes inhibited: CYP2C19

Elimination

Half-life: 0.5-1 hr; increases to 3 hr with hepatic impairment

Dialyzable: No

Total body clearance: 500-600 mL/min

Excretion: Urine (77%); feces (16-19%; mainly in bile)

Pharmacogenomics

CYP2C19 poor metabolizers

- Asians have ~4-fold higher exposure to omeprazole than whites
- CYP2C19, a polymorphic enzyme, is involved in the metabolism of omeprazole

- ~15-20% of Asians are CYP2C19 poor metabolizers
- Tests are available to identify a patient's CYP2C19 genotype
- Avoid use in Asian patients with unknown CYP2C19 genotype or those who are known to be poor metabolizers

Formulary

Adding plans allows you to compare formulary status to other drugs in the same class.

Formulary information is available for health plans only in the United States.